

Mathematics—Grade 12 Introductory Calculus with Exercises

Subject: Mathematics

Calculus studies change and accumulation. A limit describes the value a function approaches as its input approaches a particular number. A function is continuous at a point when its value exists, its limit exists, and the limit equals the function value. Continuity allows a graph to be drawn near a point without an unexpected break, although a function can still have corners or vertical tangent behavior.

The derivative measures instantaneous rate of change and gives the slope of a tangent line. For a power function, the rule is $d(x^n)/dx = nx^{n-1}$. Thus the derivative of x^4 is $4x^3$, and the derivative of $7x^2$ is $14x$. Constants have derivative zero. The sum rule allows terms to be differentiated separately. The product rule is $(fg)' = f'g + fg'$, and the chain rule is used for a function inside another function: the derivative of $f(g(x))$ is $f'(g(x))g'(x)$. Derivatives can describe velocity from position, acceleration from velocity, and marginal change in economics.

Critical points occur where $f'(x)$ is zero or undefined. A sign change in the derivative can show whether a function changes from increasing to decreasing or from decreasing to increasing. The first pattern indicates a local maximum, while the second indicates a local minimum. For an absolute maximum or minimum on a closed interval, compare function values at critical points and endpoints. A derivative can also help find tangent lines. If a curve has value $f(a)$ and slope $f'(a)$ at $x = a$, its tangent equation is $y - f(a) = f'(a)(x - a)$.

Integration reverses differentiation. An antiderivative F of f satisfies $F'(x) = f(x)$. The Fundamental Theorem of Calculus states that the definite integral from a to b of $f(x) dx$ equals $F(b) - F(a)$. A definite integral can represent signed area, accumulated distance, or total quantity produced from a rate. If velocity is positive, integrating velocity over time gives displacement; total distance may require considering direction. Units are a useful check because integrating a rate with respect to time returns the original quantity.

Exercise 1: Differentiate $f(x) = 3x^2 + 4x - 5$.

Exercise 2: Evaluate the integral from 0 to 2 of $3x^2 dx$.